XR Venue Lab – Recording Guide

VERSION 1.0 · 11/03/2026 · FULL VERSION

XR Venue Lab has up to 4 video outputs (REC OUTPUT) that send the screen, the technician view or any camera via SPOUT or NDI. OBS Studio records them: the program sets OBS up by itself, creates its own scenes without touching yours, and ● REC records what you choose (any output and/or the MULTI grid), each to its own file.

What you need

WHAT	WHERE
OBS Studio 30 or newer	obsproject.com. It includes the WebSocket XR Venue Lab uses to set it up.
Spout2 plugin (Off World Live)	Receives the SPOUT outputs (same PC).
DistroAV	Receives the NDI outputs. Needs the NDI Runtime (its installer offers it if missing).
Source Record (Exeldro)	Records each output to its own file. Without it OBS only records what is on air.

> Everything is installed from the program itself (INSTALL) or by hand from the official page (PAGE). The plugins are free and open source; XR Venue Lab only downloads their official installers from GitHub.

REC OUTPUT panel (0)

Opens and closes with 0, below the CAMERAS panel. Outputs always start off when the program opens.

BUTTON	WHAT IT DOES
XRVL OUT 1 … 4	Output name, the same you will see in OBS or Resolume. Blue NDI, violet SPOUT; on the right, the real frames per second. Red border = on.
ON / OFF	Turns the output on or off.
NDI / SPOUT	SPOUT: to OBS on this same PC (fastest). NDI: to another PC on the network, or this one.
PGM TECH 1 … 7	What goes to the output. PGM = the screen as it is (AUTO SWITCH or the camera you pick). Only one output can be PGM. Each camera view is chosen in CAMERAS.
1080 30 FPS HIGH X-RAY	Resolution, frames per second, quality (HIGH identical to the screen, LIGHT lighter) and X-RAY view. One click switches each.

OBS box

BUTTON	WHAT IT DOES
Status	not checked · not installed · WebSocket off · closed · connected (with the OBS version).
CHECK	Checks OBS, its WebSocket and the plugins.
INFO	Opens a window with a short version of this guide.
GET OBS STUDIO	Shown when OBS is not installed: opens obsproject.com.
ENABLE OBS WEBSOCKET	Shown when the WebSocket is off. With OBS closed, it turns on OBS's own remote control (only on this PC).
OPEN OBS	Opens OBS and connects by itself.
Plugins: OK / FOUND / MISSING	OK: confirmed by OBS. FOUND: installed but OBS is closed. MISSING: not installed.
INSTALL	With OBS closed: downloads the official installer and runs it (Windows asks for permission). Follow the installer, then CHECK.
PAGE	Opens the official plugin page to install it by hand.
SET UP SCENES	Creates or updates the scenes XRVL OUT 1 to 4 (one Spout or NDI source each) and XRVL MULTI (all 4 in a grid) in your current scene collection. Your own scenes and sources are not touched.
RECORD 1 2 3 4 MULTI	What REC records, in any combination: for example only MULTI, or 1 and MULTI, or all five. The chosen ones are red. A chosen output must be on (ON) to have an image; if it is off it is skipped with a warning. Choose before recording.
● REC / ■ STOP	Records what is chosen in RECORD, each to its own file; next to it, the recording time. If OBS was not connected, it connects first. STOP ends everything.
RECORDS FOLDER · GUIDE	Opens the recordings folder · opens this guide.

Step by step (first time)

1. Install OBS Studio (obsproject.com) if you do not have it.
2. In XR Venue Lab press `0`, then CHECK.
3. If it says **WebSocket off**: close OBS and press ENABLE OBS WEBSOCKET.
4. With OBS closed, press INSTALL on each plugin that says MISSING and follow each installer to the end.
5. Press OPEN OBS: it should say **connected** and the plugins **OK**.
6. Choose what goes to each output, turn on the ones you want (ON) and press SET UP SCENES. The new XRVL scenes appear in OBS.
7. In RECORD choose what to record (1 to 4 and/or MULTI), ● REC to record and ■ STOP to finish. RECORDS FOLDER opens the files.

> Next times, just open OBS (OPEN OBS), turn the outputs on and press ● REC: REC updates the scenes by itself before recording (for example if you switched an output from SPOUT to NDI).

Installing the plugins by hand

WHAT	WHERE

Spout2 plugin	github.com/Off-World-Live/obs-spout2-plugin → Releases → download the Windows installer (.exe) → close OBS → run it → Next until Finish.
DistroAV (NDI)	github.com/DistroAV/DistroAV → Releases → download the Windows installer → close OBS → run it. If it asks for the NDI Runtime, accept (or get it from ndi.video).
Source Record	obsproject.com/forum/resources/source-record.1285 → Download → pick the Windows installer → close OBS → run it.

After installing, open OBS and press CHECK in XR Venue Lab: the plugin should turn OK.

The files

- They are saved in the **Records** folder, next to the program (XRVenueLab.exe).
- Name: output + date and time it started, for example **XRVL OUT 1_20261103-170512.mkv** (year, month, day - hour, minute, second) and **XRVL MULTI_...**.
- MKV format: if OBS closes suddenly, the recording is not lost. To convert to MP4: in OBS, File → Remux Recordings.
- To change the encoder or format of a recording: in OBS, scene XRVL OUT n → Filters → XRVL REC (for example NVENC, to free the CPU).

Performance

PGM costs almost nothing (it copies the screen). Each camera in HIGH renders the venue again: about 5–6 ms of GPU per camera (RTX 3060). If FPS drop: LIGHT, fewer outputs, 720 or FPS 30. With Resolume on the same PC, set FPS 30 in the menu (GRAPHICS) and use PGM plus 1 or 2 cameras in HIGH. Recording 4 files at once also uses the disk and the GPU encoder: check each output FPS before recording something important.

If something does not work

PROBLEM	WHAT TO DO
WebSocket off	Close OBS, ENABLE OBS WEBSOCKET and open it with OPEN OBS.
"OBS rejected the WebSocket password"	In OBS: Tools → WebSocket Server Settings: leave the password as is, close OBS and press CHECK again.
A plugin is still MISSING after installing it	Close and reopen OBS (plugins load at startup) and press CHECK.
Scene XRVL OUT n is black	That output is off, or OBS still has the other mode (Spout/NDI): turn it on and press SET UP SCENES.
The NDI source does not show up	Both PCs must be on the same network and Windows Firewall must allow NDI.
No file appears	Source Record is missing, it was not chosen in RECORD, or the output was not on when you pressed REC.

Recording on another PC (NDI)

The automatic setup is for OBS on this same PC. To record on another PC on the network: set the output to NDI, install OBS and DistroAV on the other PC and add an NDI source named **THIS-PC-NAME (XRVL OUT n)**.